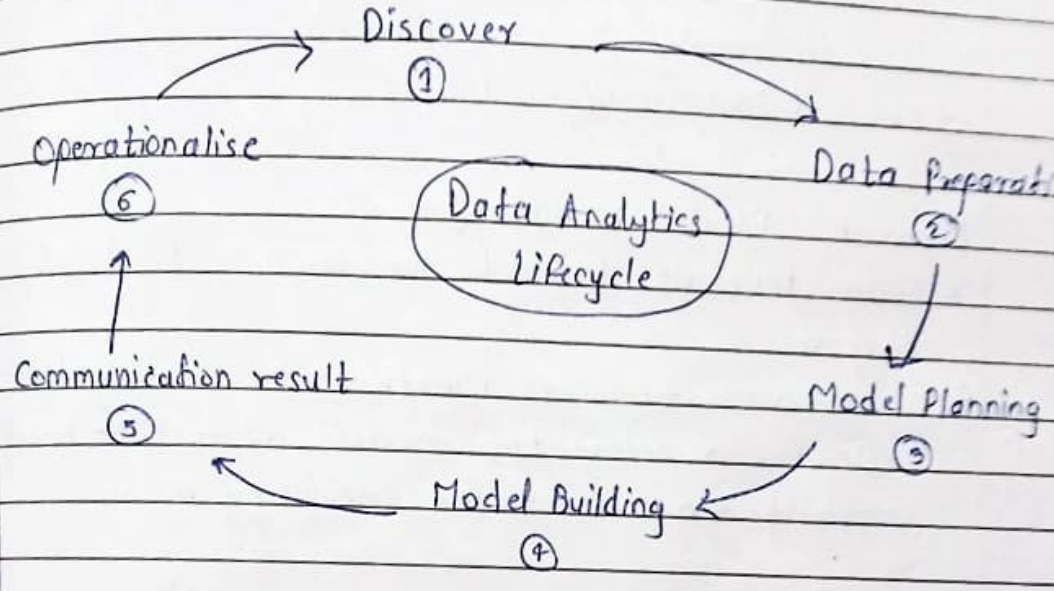
Unit - III

Predictive Data Analytics with Python

Data Analytics Lifecycle:

The Data analytic lifecycle is designed for Big Data problems and data science projects. The cycle is iterative to represent real project. To address the distinct requirements for performing analysis on Big Data, step-by-step methodology is needed to organize the activities and tasks involved with acquiring, processing analyzing data.



Phase 1: Discovery →

- The data science team learns & investigates the problem
- Develop context and understanding.
- come to know about data sources needed & available for the project
- The team formulates the initial hypothesis that can be later tested with data.

Phase 2: Data Preparation →

- steps to explore, preprocess and condition data before modeling and analysis.

- Data preparation tasks are likely to be performed multiple times and not in predefined order

Phase 3: Model Planning :→
- The team explores data to learn about relationships between variables and subsequently, selects key variables and the most suitable models.
- In this phase, the data science team develops data set for training, testing & production purposes.
- Team builds & executes model based on the work done in model planning phase.
- Several tools used - Matlab.

Phase 4: Model Building :→
- Team develops datasets for training, testing & production purposes.
- Team also considers whether its existing tools will able for running the models or if they need more robust environment for executing models.

Phase 5: Communication results →
- After executing model team need to compare outcomes of modeling to criteria established for success or failure.
- Team should identify key findings, quantify business value and develop narrative to summarize and convey findings to stakeholders.

Phase 6: Operationalize →
- The team communicates benefits of project more broadly and sets up project to deploy work in users.
- The team delivers final reports, briefings, codes.

Analytics Types:

A) Predictive Analytics: (Future prediction)

- Predictive analytics helps your organization predict with confidence what will happen next so that you can make smarter decisions and improve business outcomes
- The purpose of predictive model is finding the likelihood different samples will perform in a specific way.
- This models typically utilizes a variety of variable data to make the prediction. The variability of the component data will have a relationship with the what it is likely to predict.
- They also help forecast demand for inputs from the supply chain, operations and inventory.

Process involved

1) Project defination: All things such as outcomes, objectives etc.
2) Data Collection: to collect data from various sources
3) Analysis: Data inspection, cleaning, transformation.
4) Statistics: Finding assumptions, findings & hypothesis
5) Modelling: Through this accurate predictive models are choose
6) Deployment: Model are deployed to see reports, results & other metrics
7) Monitoring: Models are monitored to control & check for performance the results are expected or not.

e.g. Retail, weather, social media analysis, Healthcare fraud detection etc.

B) Descriptive Analytics: (Past trends)

- It is simple method and used in first phase of analytics, involves gathering, organizing, tabulating datas then the characteristics of what we are studying.

- This models shows relationships between the customer and product with the acquired data. This model can be used to organize a customer by their personal preferences for example.
- Descriptive statistics are useful to show the things like total stock in inventory, average dollars spent per user and year over year changes in sales.
- e.g.

To provide historical insights regarding the company's production, finacials, operations, sales, finance, inventory and customers.

- The main things are used in this are business intelligence tools

e.g. POWERBI, Tableau etc.

c] Prescriptive Analytics:

- This model suggests a course of action. Prescriptive analytics assists user in finding the optimal solution to a problem or in making the right choice among several alternatives.
- The mode utilizes an understanding of what has happened, why it has happened and variety of "what-might-happen" analysis to help the user determine the best course of action to take.
- This analysis is typically not just with one individual response but is, in fact, a host of other actions.
- e.g.

Traffic applications helping you choose the best route home and taking into account the distance of each route, the speed at which one can travel on each road & crucially, the current traffic constraints.

- It is mainly used for scheduling the things.
- With known parameters, prescriptive analytics not only can anticipate what will happen and when, but it also explains why it will happen. It can automatically improve prediction accuracy and inform the best next step because it can continually take in new data to re-predict & re-prescribe.

Regression →

- Regression analysis is a supervised learning analysis where we have to analyze or predict the data based on the previously available data or past data.
- It is mainly used for finding the relationship between the two variables and predicting future results based on past relationship.
- A regression task begins with a data set in which the target values are known.
- It is good choice when all of the predictor variable are continuosly valued as well.
- For an input x, if the output is continuos, this is called as regression problem.

e.g. Based on historical info of demand for oil in the market you are asked to predict the demand for next month.

- Regression is concerned with the prediction of continuos quantities. linear regression is the oldest and most widely used predictive model in field of machine learning.

- The goal is to minimize the sum of squared errors to fit a straight line to a set of data points.

* Accuracy metrics in Regression →

1) RMSE (Root mean Square Error) → Measures the difference between predicted and actual values.
↓ lesser value → Better is model ↑

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2}$$

2) MSE (Mean Squared Error) → How the regression line is close to the set of data points.
↓ lesser value or error → Better estimation ↑

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2$$

3) Mean Absolute percentage Error (MAPE) → Evaluates the prediction accuracy in percentage terms.

* Regression Line →

- Least squares : The least squares regression line is the that makes sum of squared residuals as small as possible. Linear means "Straigh line".
- Regression line is line which gives best estimate of one variable from the value of any other given variable
- It gives the average relationship between the two variables in mathematical form.

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- Regression line of X on Y → Gives the best estimate for any specific given value of Y:

$$X = a + bY$$

where,

a = X - intercept
b = slope of the line
X = dependent variable
Y = Independent variable

- Regression line of Y on X → Gives the best estimate for the value of Y for any specific given values of X:

$$Y = a + bx$$

where,

a = Y - intercept
b = slope of the line
Y = dependent variable
X = Independent variable.

★ Linear Regression:

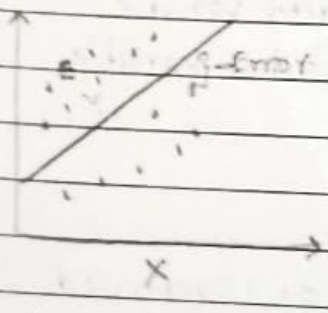
- It is also type of supervised learning that learns from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets.
- It computes the linear relationship between the dependent variable and one or more independent features by fitting a linear equation with observed data.
- It predicts the continuous output variable based on the independent input variable.

e.g. If we want to predict house price we consider various factor such as house age, distance from main road location, area and number of room, linear regression uses all these parameter to predict house price as it consider a linear relation between all these features and price of house.

- Linear regression with a single predictor can be expressed with the equation:

$$y = \theta_2 x + \theta_1 + e$$

- The regression parameters in simple linear regression are slope of the line (θ_2), the angle between a data point and the regression line & the y intercept (θ_1) the point where x crosses the y axis ($x=0$)



- Model 'Y', is a linear function of 'X'. The value of 'Y' increases or decreases in linear manner according to which the value of 'X' also changes.

* What is the best-fit line?

- Our primary objective while using linear regression is to locate the best-fit line, which implies that the error between the predicted & actual values should be kept to a minimum. There will be a least error in best-fit line.

Page No:
Date:

* Python implementation

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
# Train
model.fit(X_train, y_train)
# predict
y_pred = model.predict(X_test)
```

Logistic Regression → (classification Algorithm)

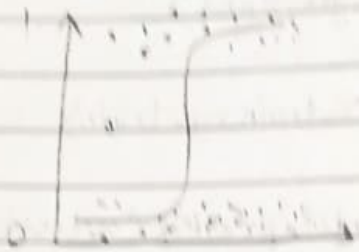
- Logistic regression is supervised machine learning algo used for classification tasks where the goal is to predict the probability that an instance belongs to a given class or not.
- Logistic regression is a statistical algorithm which analyze variables and produce a probability value between 0 and 1.
- It is used for binary classification where we use sigmoid function, that takes input as independent variables and produces a probability value bet^n 0 & 1.

e.g.
we have two classes. Class 0 and class 1, if the value of the logistic function for an input is greater than 0.5 then it belongs to class 1. otherwise it belongs to class 0.

- Logistic regression predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value.
  i.e. Target attribute is discretly changed.

$$\log\left[\frac{y}{1-1}\right] = c + B_1X_1 + B_2X_2 + \dots$$

- In logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predict two maximum value (0 or 1)



- Sigmoid function

$$f(x) = \frac{1}{1+e^{-y}} \rightarrow \frac{1}{1+e^{-(mx+c)}}$$

Three types of Logistic →

1) Binomial → 0 or 1 → two class
2) Multinomial → 3 or more classes which are unordered
3) Ordinal → 3 or more but ordered class.

Sigmoid Function →

- It is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1.
- Here we use the concept of the threshold value which defines the probability of either 0 or 1. such values above the threshold value tends to 1 & a value below the threshold values tends to 0.
  - f(x) tends to 1 as $y \rightarrow \infty$
  - f(x) tends towards 0 as $y \rightarrow -\infty$

* Evaluation of Logistic regression →

- From confusion matrix →

e.g. spam or not spam

		Predicted	
	•	spam	not spam
Actual	spam	TP	FN
	not spam	FP	TN

TP → Actual spam, predict spam

TN → Actual not spam, predict not spam

Type-I FP → Actual not-spam, predict spam

Type-II FN → Actual spam, predict not spam

From Above →

- Accuracy → It provides the proportion of correctly classified instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Precision → Precision focuses on the accuracy of positive predictions.

which is

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Recall → It measures of the proportion of correctly predicted positive instances among all actual positive instances.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- F1 Score → It is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision * Recall}{Precision + Recall}$$

→ Python implementation →

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
# Train
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Predict Probability
y_prob = model.predict_proba(X_test)
```

Naive-Bayes Classifiers →

- The Naive Bayes classifier is simple but effective probabilistic learning algorithm based on the Bayes theorem with strong independence assumptions among the features.
- It is a classification algorithm for categorical variable which is based on the well-known Bayes theorem.
- Used mostly in high-dimensional text classification.
- It is a simple probabilistic classifier and it has very few number of parameters which are used to build the ML models that can predict at a faster speed than other classification algorithm.

- e.g. It is used in spam filtration, sentimental analysis classifying articals and many more.

Naive means → where every single feature helps to identify the class without relying on other feature

Formula →

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

(Likelihood, Prior, Posterior, Marginal)

A = target → y (dependent)
B = independent $(x_1, x_2, \cdots, x_n)$

$$P(y \mid (x_1, x_2, \cdots, x_n)) = \frac{[P(x_1|y) \cdot P(x_2|y) \cdots P(x_n|y)] \cdot P(y)}{P(x_1) \cdot P(x_2) \cdot P(x_3) \cdot \cdots \cdot P(x_n)}$$

$$P(y|x_1, x_2, \cdots x_n) = \boxed{P(y) \cdot \prod_{i=1}^{n} P(x_n|y)}$$

$$\boxed{y = \underset{\uparrow}{\text{argmax}}\, P(y) \prod_{i=1}^{n} P(x_n|y)}$$

value should in range of 0 to 1
So use the Normalization.

Three types →

1) Gaussive Naive Bayes → Normal distribution, it gives bell-shaped curve.
2) Multinomial Naive Bayes → Features show Frequency, text class
3) Bernoulli Naive Bayes → Binary features.

* Python Implementation →

```
from sklearn.naive_bayes import GaussianNB
model = GaussianNB()
# Train
model.fit(X_train, y_train)
# predict
y_pred = model.predict(X_test)
```

Decision Tree →

- A decision tree is a supervised learning algorithm used for both classification and regression tasks.
- It models decisions as a tree-like structure where internal nodes represent attribute tests, branches represent attribute value and leaf nodes represent final decisions or predictions.

Internal → Test
Branches → attribute value
Leaf node → predictions or Final decision

- Information Gain →
- It tells us how useful a question is for splitting data into groups. It measures how much the uncertainty decreases after the split.
- The feature with the highest information gain is choosen to make the decision.

$$Gain(S, A) = Entropy(S) - \sum \frac{|S_v|}{S} \cdot Entropy(S_v)$$

Information Gain for all features

- Entropy : It is measure of uncertainty of random variable, it characterizes the impurity of an arbitrary collection of examples.
- The higher the entropy more the information content

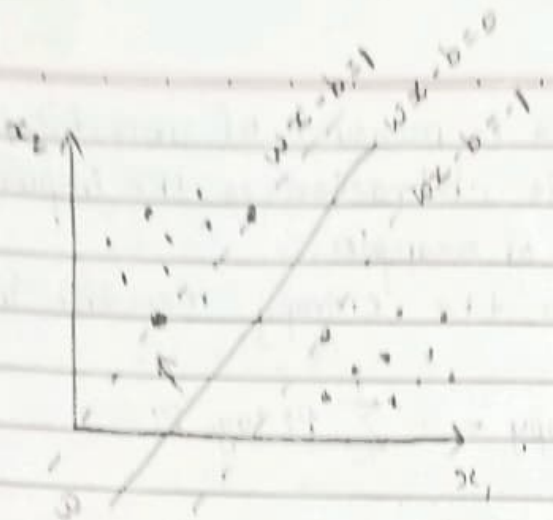
$$\text{Entropy} = -\sum_{i=1}^{n} P_i \log_2 P_i$$

* Python implementation →

```
From sklearn.tree import DecisionTreeClassifier
    model = DecisionTreeClassifier()
    # Train:
    model.fit(x_train, y_train)
    # Predict
    ypred = model.predict(x_test)
```

SVM (support vector machine)

- Support vector machine (SVM) is a supervised machine learning algorithm commonly used for classification tasks
- SVM constructs a hyperplane or set of hyperplanes in a high-dimensional space that seperates the different classes.
- A good seperation is achieved by the hyperplane that has the largest margin, meaning the maximum distance between data points of different classes.
- It is powerful classifier that works both on linearly and non-linearly seperable data. It tries to find best margin that seperates the classes.
- It is used for multiclass or binary classification



- Hyplane: This is the decision boundary represended by equation $w.x - b = 0$. It seperates the two classes. The SVM algorithm searches for the hyperplane that best divides the data while maximizing the margin.
- Support vectors: These are data points closet to the hyperplane from both classes, highlighted on the boundary lines $w.x - b = 1$ and $w.x - b = -1$. In fig. they are the points on the dashed lines representing the margin
- Margin: The margin is the distance between the support vectors and the hyperplane. The goal of SVM is to maximize this margin, ensuring that the hyperplane seperates the classes as clearly as possible
- Weight vector(w) ⇒ The arrow labeled to represents the weight vector perpendicular to the hyperplane. The direction of this vector indicates how the hyplane is oriented, and its magnitude determines how steep the slope of the seperation boundary is.

Types of SVM:

A] Linear SVM:

- In cases where data is linearly seperable, a linear SVM can find a straight line that seperates the classes

Page No.
Date.

* Python implementation →

```
from sklearn.svm import SVC
model = SVC(kernel = 'linear')
# Train
model.fit(x_train, y_train)
# Predict
y_pred = model.predict(x_test)
```

B] Non-linear SVM :→

When the data is not linearly seperable, we use ~~the~~ the non-linear SVM classifier to seperate the data points.

- we cannot draw a hyperplane.
- Here, kernel function is used to convert low-dimensional datapoints to high-dimensional data.



* Python implementation →

```
from sklearn.svm import SVC
model = SVC(kernel = 'rbf', C=1.0, gamma='scale')
# Train
model.fit(x_train, y_train)
# Predict
y_pred = model.predict(x_test)
```

1] kernel = 'rbf' → Radial Basis function ⇒ gamma is for complexity
2) kernel = 'poly' → Polynomial kernel → degree = 3, Good for medium data
3) kernel = 'sigmoid' → Similar to neural network activation function used in deep learning.

e.g. SVM

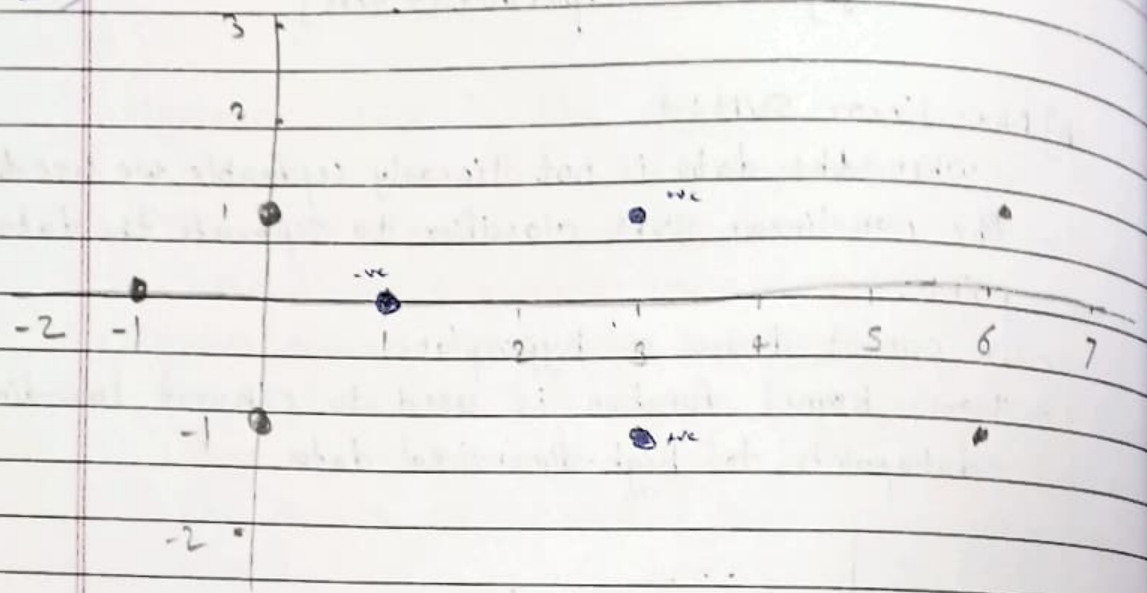
e.g. Positively labeled data points

$$\left\{ \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix} \right\}$$

& -vely labeled data points

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\}$$

→



- by inspection it should be obvious that there are 3 support vectors.

$$\left\{ s_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, s_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, s_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \right\}$$

For each vector is augmented with a 1 as a bias input

so, $s_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ then $\tilde{s_1} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

similarly

$s_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ then $\tilde{s_2} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$

$s_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ then $\tilde{s_3} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_1 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_1 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_1 = -1$$
$$\alpha_1 \tilde{S}_1 \tilde{S}_2 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_2 + \alpha_3 \tilde{S}_3 \tilde{S}_2 = +1$$
$$\alpha_1 \tilde{S}_1 \tilde{S}_3 + \alpha_3 \tilde{S}_2 \cdot \tilde{S}_3 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_3 = +1$$

$$\alpha_1 \begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix} + \alpha_2 \begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix} + \alpha_3 \begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix} + \alpha_2 \begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix} + \alpha_3 \begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix} = +1$$

$$\alpha_1 \begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix} + \alpha_2 \begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix} + \alpha_3 \begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix} = +1$$

Now,

$$\alpha_1(1+0+1) + \alpha_2(3+0+1) + \alpha_3(3+0+1) = -1$$
$$\alpha_1(3+0+1) + \alpha_2(9+1+1) + \alpha_3(9-1+1) = 1$$
$$\alpha_1(3+0+1) + \alpha_2(9-1+1) + \alpha_3(9+1+1) = 1$$

$$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$$
$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$$
$$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1$$

calculated.

$$\alpha_1 = -3.5, \quad \alpha_2 = 0.75, \quad \alpha_3 = 0.75$$

weight vector

$$\tilde{w} = \sum_i \alpha_i \tilde{S}_i$$

S - nearest data

$$= -3.5\begin{pmatrix}1\\0\\1\end{pmatrix} + 0.75\begin{pmatrix}3\\1\\1\end{pmatrix} + 0.75\begin{pmatrix}3\\-1\\1\end{pmatrix}$$

$$= \begin{pmatrix}1\\0\\-2\end{pmatrix}$$

Page No:
Date.

Finally, remembering that our vectors are augmented with bias:
- we can equate the last entry in $\tilde{w}$ as the hyperplane offset b & write the seperation
- Hyperplane equation $y = wx + b$
- with $w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ & $b = -2$